4.3 Related Rates

Name _____ Period ___ Date _____

AP Calculus 4.3 · Exit ticket

1.

$y = x^2$ and $\frac{dx}{dt} = 3$. Find $\frac{dy}{dt}$ when $x = 4$.

2.

A circle’s radius grows at 1 cm/s. Find $\frac{dA}{dt}$ when $r = 3$ cm.

3.

Why do you substitute values only after differentiating?

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